Shrinkage and Sparsity in Linear Models

A. Fixture

1 Introduction

We study the linear model in which the response depends linearly on p predictors. Least squares is the classical estimator and was analysed in detail by many authors [1, 3]. Throughout, let X denote the $N \times p$ design matrix and let y denote the response vector. The residual sum of squares is

$$\mathrm{RSS}(\beta) = \sum_{i=1}^{N} (y_i - x_i^T \beta)^2. \tag{1}$$

Minimizing (1) over β gives the normal equations, which are discussed in Section 2.

2 Ridge regression

The ridge estimator adds a penalty to the criterion in (1). Formally,

$$\hat{\beta}^{\mathrm{ridge}} = \operatorname*{argmin}_{\beta} \Big\{ \mathrm{RSS}(\beta) + \lambda \sum_{j=1}^{p} \beta_j^2 \Big\}, \tag{2}$$

where $\lambda \geq 0$ is a complexity parameter that controls the amount of shrinkage [2].

Definition 1. The *effective degrees of freedom* of the ridge fit is the trace of the hat matrix.

Lemma 1. *For every $\lambda > 0$ the minimizer in* (2) *is unique and equals $(X^T X + \lambda I)^{-1} X^T y$.*

Proof. The criterion in Eq. (2) is strictly convex when $\lambda > 0$, so its minimizer is unique. Setting the gradient to zero gives the stated formula. □

Theorem 2 (Shrinkage)**.** *The ridge coefficients shrink towards zero as λ grows. In particular, the norm of the estimator in Lemma 1 is a decreasing function of λ.*

Proof. Write the singular value decomposition $X = U D V^T$. By Lemma 1,

$$X\hat{\beta}^{\mathrm{ridge}} = \sum_{j=1}^{p} u_j \frac{d_j^2}{d_j^2 + \lambda} u_j^T y, \tag{3}$$

and each factor in (3) decreases in λ, see [3, Theorem 2]. □

Figure 1 shows the coefficient profiles. Table 1 lists degrees of freedom.

Corollary 3. *When the predictors are orthonormal, ridge regression rescales the least squares coefficients by $1/(1+\lambda)$, as* (3) *shows.*

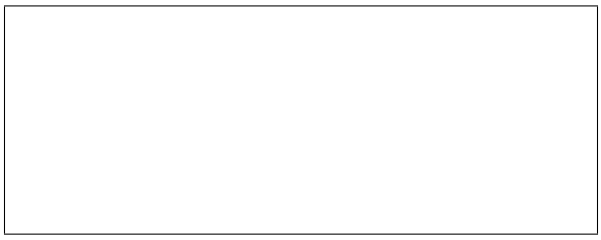

Figure 1: Profiles of ridge coefficients as the penalty λ varies.

Table 1: Effective degrees of freedom for several penalties.

λ	df
0	8.0
1	6.2
10	3.1

References

[1] A. E. Hoerl and R. W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. Technometrics, 12(1):55–67, 1970.

[2] R. Tibshirani. Regression shrinkage and selection via the lasso. Journal of the Royal Statistical Society B, 58(1):267–288, 1996.

[3] T. Hastie, R. Tibshirani, and J. Friedman. The Elements of Statistical Learning. Springer, second edition, 2009.